

Quick Check

Lesson 1: The Empty Skin on the Maize Leaf: Two Ways Insects Grow

9 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.2: Growth and Development in Animals | Lesson 1

A learner finds an empty, split insect skin on a maize leaf. A larger grasshopper is hanging beside it. Which statement describes what the learner OBSERVED?

- A** The insect peeled off its skin like sunburnt skin
- B** The skin stretched until it became the larger insect
- C** The insect melted and rebuilt itself inside the skin
- D** The empty skin is smaller than the insect beside it

In the lesson, learners are asked to write a prediction about the empty skin. At this point, before the observation stations, which response best fits the teacher's instructions for ideas such as 'it peels like sunburn'?

- A** Tell the learner at once that the idea is wrong and give the correct explanation
- B** Record the idea in the learner's own words without correcting it, and test it later with evidence
- C** Ignore the idea and move on to the next learner
- D** Remove the idea from the board because it is not scientific

At the observation stations, learners study a housefly in its stages: maggot (larva), a motionless puparium (pupa) and an adult fly. Which stage sequence did they sort for the housefly?

- A** Egg, larva, pupa, adult
- B** Egg, pupa, larva, adult
- C** Egg, nymph, nymph, adult
- D** Larva, egg, adult, pupa

Learners watch a young grasshopper nymph with a hand lens and compare it with an adult grasshopper. Which observation did the lesson lead them to record?

- A** The nymph looks like a small adult
- B** The nymph has no legs, like a maggot
- C** The nymph is a motionless resting stage
- D** The nymph looks nothing like the adult

A group sorts cards for a grasshopper and puts a pupa card between the nymph and adult cards. Which question should the teacher ask, as in the lesson?

- A** Why did you get this wrong?
- B** What evidence made you put this card here?
- C** Who in the group made this mistake?
- D** Can you copy the answer from another group?

A group has these observations: the housefly has a motionless pupa stage, but the grasshopper has no pupa stage. Which claim do these observations best support?

- A** All insects grow smoothly without changing shape
- B** Insects follow two developmental pathways
- C** All insects have exactly four life stages
- D** Insects grow by peeling like sunburnt skin

How many stages are in the life cycle of a grasshopper (incomplete metamorphosis), and how many are in the life cycle of a housefly (complete metamorphosis)?

- A** Grasshopper 3 stages, housefly 4 stages
- B** Grasshopper 4 stages, housefly 4 stages
- C** Grasshopper 3 stages, housefly 3 stages
- D** Grasshopper 4 stages, housefly 3 stages

Which question about the empty skin is the best kind of question to post on the Driving Question Board (DQB)?

A Is the skin on a maize leaf?

B Did the insect leave the skin?

C Is the skin crunchy?

D How does an insect become bigger than its old hard skin?

At the end of the lesson, which question is still NOT explained and is left for the next lessons?

- A** Which insect has a larva stage?
- B** How many stages does a grasshopper have?
- C** Why does a grasshopper grow in sudden jumps instead of a little each day?
- D** Does a housefly have a pupa stage?

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).